Name: _____

Series tests for convergence

Determine if the following series converge or diverge. If applicable, determine if the series are absolutely convergent, conditionally convergent, or divergent (this is applicable if you have some $a_n < 0$. If possible, determine what the series converges to.)

1. $\sum_{m=1}^{\infty} \frac{1}{m^2}$

2. $\sum_{n=1}^{\infty} \frac{1}{2^n}$

3. $\sum_{q=1}^{\infty} (-3)^q$

4. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{3n+1}\right)$

5. $\sum_{k=1}^{\infty} \frac{k^{2k}}{\left(1+2k^2\right)^k}$

6. $\sum_{m=1}^{\infty} \frac{m^3}{5^m}$

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7. $\sum_{j=1}^{\infty} (-1)^j \frac{1}{j+2}$

8. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n+1}$

9. $\sum_{r=1}^{\infty} (-1)^{r+1}$

10. $\sum_{b=1}^{\infty} \frac{b}{b^3+5}$

11. $\sum_{k=1}^{\infty} \frac{k^3-k+1}{\sqrt[3]{k^9+k^5}}$

12. $\sum_{j=1}^{\infty} \frac{\cos(3j)}{(1+1.2)^j}$

13. $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$

14. $\sum_{k=1}^{\infty} \frac{(-1)^k(k+1)3^k}{(k+2)!}$